

Referee Report

S.U.R.E. (Aquaculture Welfare Integration) — Area-Chair-Level Quality Gate

pAI/MSc Autonomous Research Pipeline — Reviewer

Summary

The manuscript reports a deployed sturgeon-welfare monitoring system (S.U.R.E.) in which a locally hosted LLM (AQUA-1B) proposes a severity label and a deterministic rule engine holds escalation-only final authority. Against a naive expectation that the rule engine’s value comes from correcting nuanced model misjudgments, the paper’s central, honestly reported finding is that this is not what happens: across the evaluation harness’s entire fixed scenario population ($n = 8$), the dominant pathway is fail-safe defaulting on unparseable model output (50%), not sophisticated correction (12%); and even in the “agrees” bucket, two of three cases reach the correct status label via a free-text reasoning chain that fabricates a sensor value never present in the input. Four further measurements (agentic tool-routing, RAG threshold calibration, PSI drift-binning, and vision operating-point calibration) extend the same narrow-enumerable-interface argument across architecturally distinct layers of one production system, including one full disclosed negative result and one live, unresolved production data-integrity bug (a stale 0.695 recall figure inside the deployed RAG knowledge base) that the paper’s own audit discipline caught.

This is a well-executed, honestly self-critical applied-systems paper. Eleven experiments were independently re-run and verified against raw output; eight of them were re-checked directly against their source `results.md`/JSON artifacts during this review, and all eight matched the manuscript’s reported numbers exactly (Section 2 below). The Related Work section correctly declines to claim invention of deterministic override, interlocks, or shielding, naming IEC 61508/61511 Safety Instrumented Systems and RL-safety shielding explicitly; an independent web search (Section 3 below) confirms this framing is accurate. No hard blocker (B1–B5) is triggered, and the AI-voice audit independently confirms the proofreader’s finding of zero violations. One concrete, easily-fixed factual inconsistency was found in the Introduction (Section 4 below) and should be corrected before publication, but it does not touch any headline scientific claim.

Recommendation: Accept with minor revisions (one required text fix).

Strengths

- **Claim traceability.** Every numeric claim spot-checked against `formalized_results.json` and the underlying `experiment_runs/EXPnn/results.md` files reproduced exactly: EXP01’s RAG F1-argmax (0.84, F1=0.951) and random-baseline margins (hit@1 +0.586, MRR +0.438); EXP02’s 8/8 module-identity check; EXP03’s four-bucket distribution (50/38/12/0%) and the 3-of-8 fabrication count; EXP04’s 0%/0% AQUA-1B result and AQUA-7B’s constant-answer artifact at $n = 9$; EXP05’s corrected vision range (P 0.846–0.852 / R 0.712 / mAP50 0.8335–0.8345); EXP06’s INT8-loses-less-than-fp32 finding (−0.0082 vs. −0.0104) and CoreML/ANE latency (9.03/9.60 ms); EXP07’s deployed-vs-argmax recall split (0.782 vs. 0.719); and EXP08’s

PSI crossover values (0.164 vs. 0.078 at $\delta=0.02$; 4.844 vs. 7.649 at $\delta=0.30$). No fabricated or rounded-away numbers were found.

- **Honest handling of the central $n = 8$ result.** The Abstract, Introduction, Results, Discussion, and Known Limitations all state the sample size explicitly, disclaim any implied confidence interval, and frame the finding as a census of a fixed scenario population rather than a statistically powered estimate. The adapter-provenance caveat (the deployed adapter is an 8-example, self-documented-as-insufficient smoke-test artifact) is scoped correctly throughout – it is never misapplied to the agentic-routing result (Results 6.2, which ran the base model with no adapter loaded), and the paper explicitly checked this and stated the distinction (Known Limitations, ‘The adapter-provenance caveat is scoped narrowly’).
- **Prominent disclosure of the live RAG leak.** The stale 0.695 recall figure inside the production knowledge base is disclosed in five separate, prominent locations: a full rhetorical paragraph in the Introduction, the Section 6.6 narrative, Table 3 (severity-ranked, row 1, marked “not fixed”), Known Limitations, and Discussion 7.3’s “fifth pattern” paragraph. This is not a footnote-buried admission; a reader skimming any one of the paper’s major sections would encounter it. This reviewer independently verified against the live git history of the underlying repository that the figure is (a) real – commit 0e8b414 introduced the stale value and 9a260af corrected the figure elsewhere 43 minutes later without touching this file – and (b) still unfixed at the time of this review.
- **Accurate, non-inflated novelty positioning.** Section 2.2 (“What this paper does not claim to have invented”) explicitly cites SIS/IEC 61508–61511 and RL-safety shielding and states plainly that deterministic override of a probabilistic component is decades-old prior art. An independent web search confirms both characterizations: SIS/IPL literature does describe a diagnosed, non-bypassable final-authority layer built around enumerable hazards, and the RL shielding literature does require a formally specifiable safety condition with guarantees that hold only within that specification – supporting the paper’s claim that neither literature has a direct equivalent for a fabricated free-text justification riding alongside a correct enumerable decision.
- **Genuinely low AI-voice risk.** No Furthermore/Moreover/Additionally chains exist (only three isolated, non-chained uses of “additionally” in the whole 31-page manuscript); Related Work is organized as a critical, opinionated two-cluster-plus-pivot argument, not a flat catalogue; sentence length varies naturally within paragraphs; lists are stated at their true size (four practitioner rules, five takeaways) rather than padded or trimmed to three; the Conclusion adds new synthesis rather than re-abstracting the Abstract.

Weaknesses

- **An internal factual inconsistency about the RAG-leak timeline.** The Introduction states the stale knowledge-base figure was found “*weeks after the correction*” (`sections/introduction.tex`, line 71). This directly contradicts Section 6.6 (`results.tex`, line 350) and Known Limitations (`known_limitations.tex`, line 44), both of which correctly state the gap as “43 minutes’ worth of commits.” This reviewer verified against the actual git log that the two commits in question (0e8b414 and 9a260af) are 43 minutes apart on the same day – the Introduction’s “weeks” framing is not just inconsistent with the rest of the paper, it is factually wrong. This is a minor, local, one-clause error with no bearing on any headline scientific claim, but it sits

inside the paper’s own showcase paragraph about catching stale claims through disciplined auditing, which makes the inconsistency more conspicuous than its substance warrants. See Must-Fix Action 1.

- **Contribution is real but modest, and the paper is right to say so.** The literature review’s own `novelty_flags.json` classifies the core architectural pattern as `EQUIVALENT_KNOWN`; the paper’s defensible novelty locus is the cross-layer measurement plus the fabrication-detection finding, not the architecture itself. The paper does not overclaim here, but a reader expecting a new mechanism rather than a rigorously measured deployment of a known one should have that expectation set correctly by the Abstract – which it is, but only implicitly (the Abstract never states outright that deterministic override itself is known prior art; a reader who skips Related Work could momentarily overestimate the architectural novelty before reaching Section 2.2).
- **Numeric density in Results is high.** The copy-editing pass already reviewed this and found every number individually source-cited and load-bearing; this reviewer agrees no paragraph should be cut, but flags for the record that Sections 6.3–6.6 in particular ask a lot of a reader tracking several near-identical-looking numbers (0.719 vs. 0.782 vs. 0.720 vs. 0.858 vs. 0.859) across a short span of text. This is a concision trade-off the paper makes deliberately and discloses (Table 2 exists precisely to consolidate this), so it is noted as a nice-to-fix, not a required change.
- **The live production leak remains unfixed as of this review.** This is explicitly disclosed as an intentional consequence of the audit’s read-only guardrail, not an oversight, and the manuscript is honest about it. It is noted here only because a reader may reasonably ask why a team that found this would not simply fix the one-line knowledge-base file before submission; the paper’s own rationale (a hard read-only constraint during the audit) is reasonable but could be stated a half-sentence more explicitly next to the disclosure itself, rather than relying on the reader to infer it from “consistent with a read-only guardrail on the live system.”

Detailed Comments

Hard blockers (B1–B5): none triggered. **B1/B2** (research questions + evidence-pointing takeaways): the Introduction states three explicit `\question{}` blocks and five explicit `\takeaway{}` boxes, each naming a specific section and, where applicable, a specific figure. All five takeaways were checked against their cited section and found supported. **B3** (placeholders): a manuscript-wide grep for `TODO`, `TBD`, `[cite:, ??, XXX, FIXME` across `final_paper.tex` and every `sections/*.tex` file returned zero matches; the compile log shows zero undefined references or citations. **B4** (repetition/filler/templated feel): not triggered – see AI-voice discussion below. **B5** (unproved theorem claims): confirmed N/A. `theory_track_summary.json` is intentionally empty (empirical-only project, per `track_decomposition.json`’s `recommended_track=empirical`). All four Formalizations in Section 4 are explicitly and correctly labeled as restatements of already-implemented code, not proofs – Formalization 1’s own text states outright, “This is not a proved theorem; it is a restatement, in one line, of what `apply_rule_override` already does in code.” No section anywhere claims a Formalization is “proved” or attains theorem-status.

AI-voice risk: LOW (independently confirmed, not taken on faith from the copyedit report). Checked against the concrete criteria: the Abstract contains no citations or `\ref{}`

references and does not use “Our contributions are N-fold”; a manuscript-wide grep found zero Furthermore/Moreover chains and only three isolated instances of “additionally,” none in a repeated pattern; Related Work is a critical, two-cluster-plus-pivot argument with explicit judgment (“the honest answer to the second question is no, and stating that plainly... is more useful to a reader than pretending otherwise”), not an enumerative catalogue; the Conclusion introduces new synthesis (the SIS/shielding comparison, the “what would strengthen this claim further” paragraph) rather than restating the Abstract; sentence length and paragraph structure vary naturally throughout; the practitioner-rule list in Discussion 7.3 has four items and the Introduction’s takeaway list has five, neither trimmed or padded to a suspicious round three.

$n = 8$ overreach check (Abstract/Introduction specifically). Neither the Abstract nor the Introduction implies a confidence interval, a rate that would hold over a larger population, or generalization beyond the eight evaluation-harness scenarios. The Abstract’s language (“genuine correction occurs in only one case in eight”) states the count as a fraction of eight explicitly rather than as a percentage that could read as a population estimate. The one generalization claim in the entire paper (Discussion 7.4, “these results suggest, but do not demonstrate beyond this one system...”) is explicitly hedged, correctly scoped to the cross-layer consistency argument (which rests on five independent layers, not on $n = 8$ alone), and uses “suggest”/“may point to” rather than “prove” or “show,” per the author style guide’s own binding instruction.

Numeric spot-checks (8 performed, all matched). See Strengths, first bullet, for the full list of experiments checked directly against `experiment_workspace/experiment_runs/ EXPnn/results.md` and `formalized_results.json`.

Novelty spot-check (WebSearch performed). Confirmed IEC 61508/61511 Safety Instrumented Systems are described in the standards literature as a diagnosed, non-bypassable final-authority layer for enumerable process hazards (Independent Protection Layer framing), matching Section 2.2’s characterization. Confirmed the RL-safety shielding literature builds shields from a formal specification of safety-critical properties with guarantees that hold only within that specification – consistent with the paper’s claim that shielding “requires a formally specifiable, enumerable safety condition and offers no guarantee once the thing being checked is open-ended.” The paper’s claim that neither literature has a direct equivalent for fabrication-inside-a-correct-decision is a reasonable, defensible reading of this search’s results and is not an overclaim.

Questions for Authors

1. Section 6.6/Known Limitations state the RAG-leak gap as “43 minutes,” but the Introduction says “weeks after the correction.” Which is intended, and can the Introduction sentence be reconciled with the other two occurrences before the next revision?
2. The Abstract states the architecture “can be made safe against an unreliable model provided its validator reads nothing beyond one enumerable field.” Given Formalization 2’s own scope note that this protects only against *wrong enumerable decisions*, not against a fabricated free-text justification attached to a *correct* one – would it strengthen the Abstract to make that scope boundary one clause more explicit, given that EF1 (fabrication inside the “agrees” bucket) is arguably the paper’s most novel single finding?

3. Given that the live 0.695 knowledge-base leak is still unfixed as of this review, is there a planned commit to correct `llm-service/knowledge/06-davranis-ve-refah-gostergeleri.md` before or shortly after publication, and would the authors consider a one-line addendum once it is fixed (the paper’s own `followup_recommendations` already names this as the first recommended action)?

Recommendation

Accept with minor revisions. This is a rigorous, honestly self-critical applied-systems paper with fully traceable claims, an accurate and non-inflated novelty position, and no hard blockers. The one required fix (Must-Fix Action 1 below) is a single-sentence correction with no experimental or theoretical work attached. No experiment needs to be re-run and no claim needs to be walked back.